A Context-Integrated Transformer-Based Neural Network for Auction Design

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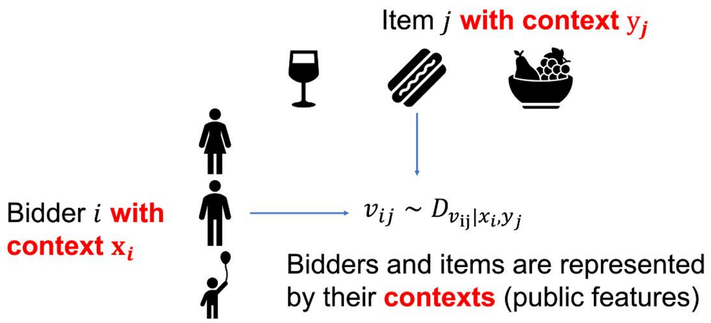
Motivation

- Pioneered by RegretNet[1], there has been much progress on on finding (approximate) optimal auction through deep learning.
- In practice, auctions are much complex: there are usually a large number of bidders and items with various features, and each auction involves a different number of bidders and items.
- we need a new architecture that can incorporate public features (contexts) and take a different number of bidders and items as inputs.

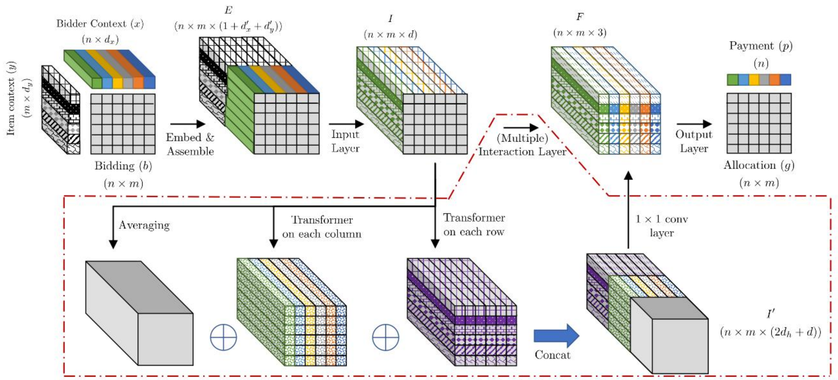
Contextual Auction



The contexts can characterize various bidders and items to some extent, making the auctions close to those in practice.

[1] Dütting, Paul, et al. "Optimal auctions through deep learning." International Conference on Machine Learning. PMLR, 2019.

Architecture: CITransNet



- Input Layer ⟹ Obtain initial representations.
- (Multiple) Interaction Layers ⟹ Model the interactions between bidders and items.
- Output Layer ⟹ Allocation and payment results.

Learning Framework

- formalize as a constraint optimization:

$$\min_{(g,p)\in\mathcal{M}} -\mathbb{E}_{(v,x,y)\sim\mathcal{D}_{v,x,y}}\left[\sum_{i=1}^{n} p_i(v,x,y)\right]$$

$$\text{s.t. } \mathbb{E}_{(v,x,y)\sim\mathcal{D}_{v,x,y}}\left[\sum_{i=1}^{n} rgt_i(v,x,y)\right] = 0,$$

- ➢ Objective: maximize the expected revenue
- ➢ Constraint: DSIC (Dominant Strategy Incentive Compatible) condition.

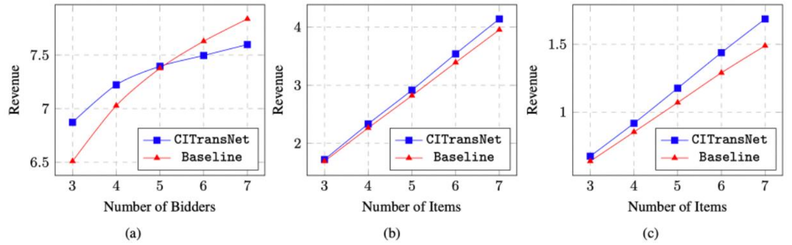
- Parameterize the auction mechanism, and find near-optimal solutions using machine learning:
  - ➢ All the expectation terms are computed empirically
  - ➢ Optimized through the augmented Lagrangian method and gradient descent.

Experiments

1. Single-item contextual auctions

Method	A: 3×1, $\lvert\mathcal{X}\rvert = 5, \lvert\mathcal{Y}\rvert = 1$		B: 3×1, $\lvert\mathcal{X}\rvert = 5, \lvert\mathcal{Y}\rvert = 2$		C: 5×1, $\mathcal{X}, \mathcal{Y} \subset \mathbb{R}^{10}$	
	rev	*rgt*	*rev*	*rgt*	*rev*	*rgt*
Optimal	0.594	-	0.456	-	0.367	-
RegretNet	0.516	<0.001	0.412	<0.001	0.329	<0.001
EquivariantNet	0.498	<0.001	0.403	<0.001	0.311	<0.001
CIRegretNet	0.594	<0.001	0.453	<0.001	0.364	<0.001
CIEquivariantNet	0.590	<0.001	0.452	<0.001	0.360	<0.001
CITransNet	0.593	<0.001	0.454	<0.001	0.366	<0.001

2. Multi-item contextual auctions

Method	D: 2×5, $\lvert\mathcal{X}\rvert = \lvert\mathcal{Y}\rvert = 10$		E: 3×10, $\lvert\mathcal{X}\rvert = \lvert\mathcal{Y}\rvert = 10$		F: 5×10, $\lvert\mathcal{X}\rvert = \lvert\mathcal{Y}\rvert = 10$		G: 2×5, $\mathcal{X}, \mathcal{Y} \subset \mathbb{R}^{10}$		H: 3×10, $\mathcal{X}, \mathcal{Y} \subset \mathbb{R}^{10}$		I: 5×10, $\mathcal{X}, \mathcal{Y} \subset \mathbb{R}^{10}$	
	rev	*rgt*	*rev*	*rgt*	*rev*	*rgt*	*rev*	*rgt*	*rev*	*rgt*	*rev*	*rgt*
Item-wise Myerson	2.821	-	6.509	-	7.376	-	1.071	-	2.793	-	3.684	-
CIRegretNet	2.803	<0.001	5.846	<0.001	6.339	<0.003	1.104	<0.001	2.424	<0.001	2.999	<0.001
CIEquivariantNet	2.841	<0.001	6.703	<0.001	7.602	<0.003	1.147	<0.001	2.872	<0.001	3.806	<0.001
CITransNet	**2.916**	<0.001	**6.872**	<0.001	**7.778**	<0.003	**1.177**	<0.001	**2.918**	<0.001	**3.899**	<0.001

3. Out-of-setting Generalization